Juan David Muñoz Bolaños

Optical Systems Engineer

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Summary

Physicist and optical engineer wrapping up a PhD in June 2026. Four years spent building, aligning and commissioning complex laser- and light-based measurement systems from scratch, hardware and software both. I integrate lasers, scanners, cameras and sensors, drive them with my own code, and when something doesn't work I trace it across optics, mechanics, electrics and software until I find the cause. Earlier, a year in the field installing and adapting optical inspection machines at customer sites. Comfortable bridging deep optics with hands-on integration and customer work. Based in Tirol, cleared to work in Austria, ready to relocate to Vienna.

Research/Professional Experience

Jan 2022 to Jun 2026 **Med. Univ. Innsbruck** *Innsbruck, Austria*
Optical Systems Engineer | PhD Researcher

Laser, SLM & DMD-Based System Build: Built complex laser- and light-based systems from the bare table up: integrated laser sources, spatial light modulators (SLMs), Texas Instruments DMDs and a TI DLP development kit for image projection, galvo scanners, scientific cameras and piezo stages. Did the alignment, calibration and optical fault-finding myself, programming devices both via microcontroller and at the projection level.

Hardware-Software Integration & Commissioning: Wrote the control and acquisition software in LabVIEW (Real Time, FPGA), Python and C++. Brought each system from first light to repeatable, calibrated operation, and traced faults across optics, mechanics, electrics and software instead of guessing.

Specifications & Documentation: Wrote the test protocols and technical specs for setup, calibration and routine operation. Trained new students and interns as their first technical contact on the rig.

Mar 2018 to Feb 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer (Field Service)

Customer-Site Implementation: Installed, commissioned and adapted automated optical inspection machines (Halcon, Cognex VisionPro) directly at customer production sites. Tuned each system to the specific application and got customer sign-off before leaving.

Support, Training & Root Cause Analysis: Handled on-site adjustments, repairs and end-user training. Worked with production and maintenance teams, ran root cause analysis on live lines, and pushed through the fixes that made the machines more reliable.

Technical Competencies

Optics, Laser & Light Modulation: Laser integration and alignment, spatial light modulators (SLMs), Texas Instruments DMDs and DLP development kits for image projection, illumination design, interferometry, wavefront sensing, ZEMAX/OpticStudio, galvo scanners, fiber optics.

System Integration & Software: Hardware-software integration, Python (NumPy, SciPy, OpenCV), C++, LabVIEW (Real Time, FPGA), MATLAB, DAQ cards (National Instruments), microcontrollers (Arduino), Git.

Machine Vision & Inspection: Halcon, Cognex VisionPro, OpenCV, automated defect detection, industrial cameras (Basler, Andor, Hamamatsu).

Hands-on & Mechanical: Opto-mechanical design (SolidWorks, FreeCAD), workshop assembly and alignment, calibration, rapid prototyping (3D printing), piezo stages (PI).

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

2019 to 2021 **Friedrich Schiller Univ. Jena** *Jena, Germany*
Master of Science in Photonics

2012 to 2018 **University of Cauca** *Popayán, Colombia*
Bachelor of Science in Physics Engineering

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Self-Driven & Adaptable: I like taking an idea and getting it to actually work. Comfortable in fast-moving environments where the plan changes mid-week.

Technical Contact & Communication: Used to working across electrical, mechanical, software and production teams. When something breaks, I dig into the cause systematically rather than guessing.

Personal Interests

• Mountain Sports (Biking, Hiking, Skiing) • Bachata Dancing • Harmonica • Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
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PD Dr. Christoph Krafft
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